Post-Training Typed-Decision Models with Proper-Scoring Losses: Accuracy, Calibration, and Soft-Target Underconfidence

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Abstract

Typed-decision models answer choice, score, and yes/no questions about a state in one forward pass and report a probability distribution that downstream thresholds consume. We post-train such models (Kev, a prefill-only pointer-head reconstruction of TypeSafe's Jev) with cross-entropy against soft gold distributions plus Brier and ranked-probability-score terms, at two scales and two data regimes. On 400 held-out business-decision cases, a 4B delta raises decision accuracy from 0.665 to 0.800 and cuts KL from gold threefold (0.226 to 0.079), while ECE regresses from 0.091 to 0.162. A cold-started 9B replication reproduces the pattern (0.774 accuracy, ECE 0.149), and a 9B model trained on a 13.4k-record mixture (v2) reaches 0.783. The ECE regression is underconfidence in every confidence bin and every question type: the expected signature of matching soft targets and then scoring against discrete labels. A single temperature below one, fit group-disjointly, restores ECE (4B: 0.009; 9B: 0.021) with accuracy unchanged, at a large cost in fidelity to the gold distribution (KL 0.079 to 0.469 at 4B). The sign of miscalibration is a property of the evaluation distribution, not of the model: the same v2 model is overconfident (bias +0.107) on 6,264 external hard-label decisions, so the in-domain temperature must not be shipped as a global fix.

1 Introduction

Autonomous systems increasingly act on model outputs without a human in the loop: routing tickets, triaging incidents, approving invoices. Each act is a typed decision, meaning a choice among named options, a score on an ordered scale, or a yes/no verdict, reported as a probability distribution. Downstream thresholds (automate, defer, escalate) are only meaningful if 0.80 means right 80% of the time. A family of single-pass "System One" decision models now targets this interface: the hosted Jev, open reconstructions such as Kev [14], Laya [3] and AnyJev [23], and domain models such as SearchJev [2]. How to post-train them so that the reported distribution is both accurate and calibrated is an open, practical question.

The recipe space is now crowded. RLCR [4] shows that correctness plus a bounded proper scoring rule incentivises calibrated verbalised confidence. SearchJev [2] trains Qwen3.5 LoRA decision models on soft targets with cross-entropy plus a Brier term, then fits a temperature per output type. Laya [3] post-trains with proper-scoring RL rewards, including RPS for ordinal questions. What these reports do not provide is a diagnosis: in which direction the resulting model is miscalibrated, on which question types, how that changes under distribution shift, and what post-hoc correction costs in fidelity to the gold distribution. Those are the quantities a deployment needs to set thresholds, and they are what this paper measures.

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Running example (invoice_processing_000008, held out). Invoice INV-2026-6038 ($36,720, Acme Fabrication) against PO-4946; delivery accepted with exceptions, received quantity 0. Gold vs. 4B delta: `discrepancy_severity` (score): gold {3:0.83} vs. model {3:0.62, 2:0.30}; `disposition` (choice): gold {manual_review:0.57, hold:0.34} vs. model {manual_review:0.50, hold:0.43}; `duplicate` (noul): gold {false:0.96} vs. model {false:0.95}; `matches_order` (noul): gold {false:0.92} vs. model {false:0.89}; `urgency` (score): gold {2:0.50, 1:0.37} vs. model {2:0.56, 1:0.27}. All five argmax decisions are correct, and the predicted mass tracks the gold mass with slightly wider spread.

Figure 1: A real held-out case: gold distributions and measured 4B-delta outputs. Section 3 uses its `urgency` question.

We post-train Kev models with soft-target cross-entropy plus Brier and RPS terms and evaluate them on a fixed typed-decision benchmark with distributional gold. Our contributions are empirical:

- Proper-loss post-training improves accuracy, proper scores and selective prediction together, at 4B (0.665 to 0.800) and in a cold-started 9B replication (0.774), with gains in every workflow and question type (Section 5).
- The resulting miscalibration is underconfidence, in every bin and every primitive, worst on ordinal score questions (ECE 0.212). This is the predicted signature of matching soft targets and then scoring against majority labels [1, 11]. The released Kev checkpoints, trained on hard labels, ship temperatures above two [14]; our deltas from the same family need $T \approx 0.4$. An independently trained Laya specialist on the same split is underconfident too (Section 6).
- One temperature restores ECE, with an explicit fidelity cost and no transfer guarantee. Group-disjoint temperatures bring ECE to 0.009 (4B) and 0.021 (9B) with accuracy fixed, but triple-to-sextuple KL from gold. On external hard-label data, the v2 model is overconfident, so the in-domain $T < 1$ points the wrong way (Section 5).

2 Background and Related Work

System One decision models. TypeSafe's Jev answers typed questions in one pass and is post-trained with an undisclosed procedure called RLCD [19]. Kev [14] is an open reconstruction: a causal LM run prefill-only, LoRA adapters, and a pointer head that scores option-boundary tokens, trained with cross-entropy on hard-label decision suites and shipped with a built-in temperature ($T = 2.41$ at 4B, 2.30 at 9B). AnyJev [23] reads decisions from an untrained LM in one prefill. Laya [3] post-trains with proper-scoring RL rewards (log and spherical, plus RPS for ordinal questions). SearchJev [2] is closest to our recipe: it fits soft target distributions with cross-entropy plus 0.5·Brier on Qwen3.5 LoRA models and fits per-output-type temperatures on 5,000 validation decisions. It reports neither the fitted temperatures nor the direction of miscalibration. Evaluations of this family compare it with fine-tuned encoders and generative LLMs [15] and report finite-sample noise floors for ECE [16]; TypedBench finds Jev itself underconfident after refitting.

Calibration under soft supervision. Modern networks trained with cross-entropy are typically overconfident and are repaired by temperature scaling [10, 6]. Softened targets push the other way. Label smoothing and soft-label augmentation produce underconfidence that compounds with other regularisers [12, 21], and models trained with such regularisation are harder to recalibrate post hoc [20]. When the gold itself is a distribution of human or teacher judgments, scoring calibration against the majority label penalises a model that correctly reports the disagreement [1, 13]. On ChaosNLI, soft-label training lowers KL and raises majority-label ECE, exactly our pattern [11]. Temperatures fit on discrete labels are biased towards understating annotator uncertainty [18, 7]. Our results should therefore be read as a measurement of a predicted effect in a new model family, not as the discovery of a new phenomenon.

RL and proper scores. RLCR [4] proves that correctness plus a bounded proper rule (Brier, not log loss) incentivises calibrated verbalised confidence. Follow-ups place confidence rewards on an overconfidence–underconfidence spectrum [17]. Verifiable-reward RL produces overconfident decision tokens that inherit calibration from reasoning traces [22]. Our models have no reasoning trace and report the decision distribution itself as confidence. After temperature scaling, proper-score objectives are not clearly better calibrated than cross-entropy [5], which matches our finding that post-hoc temperature, not the loss, is what sets ECE.

3 Method

A case is a state s (JSON) and questions q, each with k_q options and a soft gold distribution g obtained by averaging teacher judgments; y is its argmax. The model is a Kev pointer head: logits $z \in \mathbb{R}^{k_q}$ at option-boundary tokens of a prefill-only causal LM, $p = \mathrm{softmax}(z)$. Training minimises, per decision,

$$\mathcal{L} = \mathrm{CE}(p, g) + 0.5 \cdot \mathrm{Brier}(p, y) + 0.5 \cdot \mathrm{RPS}(p, y),$$

with the RPS term [8] only on score questions. Both auxiliary terms are strictly proper [9] and are scored against the realised label y, whereas SearchJev's Brier term is scored against g. Adaptation is LoRA (rank 16, $\alpha = 32$, all projections) plus the pointer head.

For `urgency` in Figure 1 (gold $\{2{:}0.50, 1{:}0.37, 3{:}0.10, 0{:}0.03\}$), a hard-label trainer pushes all mass to level 2. The soft cross-entropy term instead penalises starving level 1 of its 0.37, and the measured output $\{2{:}0.56, 1{:}0.27, 3{:}0.14, 0{:}0.03\}$ tracks the gold spread. That spread is what the loss asks for, and it is also why argmax confidence (0.56) understates argmax accuracy on the test set.

Temperature. Inference reports $\mathrm{softmax}(z/T)$, which leaves the argmax unchanged. We report two fits, both minimising NLL against y. (i) Train-fit: T on the model's own training split, on a 0.05 grid (4B only; $T^\star = 0.35$). The model has seen these rows, so this fit is in-distribution by construction. (ii) Cross-fit: five-fold group-disjoint fits on the test set, where each fold's T comes from the other four folds and all questions of a case stay in one fold (Kev's `out_of_fold_rows`, 121-point log grid on $[0.25, 4]$). Cross-fit is the honest estimate of what a held-out calibration set from the same distribution would give.

4 Experimental Setup

Data. v1: four business-decision workflows (agent-trace observability, customer service, invoice processing, security incidents); 1,200 training cases (6,000 decisions) and 400 held-out test cases (2,000 decisions, 1,945 scored after context rejects; 778 score, 578 choice, 589 noul). Gold distributions average teacher judgments (mean entropy 0.75 nats). Rare labels have fewer than 15 training occurrences (e.g. `action=stop`), which bounds tail-calibration claims. v2: 13,423 training records adding HelpSteer2 (score), UltraFeedback-binarised, MMLU auxiliary-train, HellaSwag and ANLI-R1 (choice), grouped-soft hate-speech ratings (score) and PubMedQA (noul) to v1. The v2 evaluation set has 6,264 scored decisions: a 12% per-source hold-out plus ChaosNLI [13] and RewardBench, kept evaluation-only (5,558 choice, 600 score, 106 noul). A normalised exact-match audit against the v1 test found no leaks.

Models. 4B delta: warm start from the released kev-4b, two epochs on v1. 9B delta: cold start from Qwen3.5-9B-Base (no trusted 9B adapter to warm-start from), same recipe. v2-9B: warm start from the 9B delta, two epochs on v2. Shared hyperparameters: lr 2×10^{-5}, batch 1×8 accumulation, gradient checkpointing, no option permutation, seed 0, one fixed configuration with no search. Compute: one RTX 3090 for v1 (about 2 h at 4B and 75 min at 9B), one A6000 for v2 (about 20 h).

Baselines. The eight released and trained Kev checkpoints (0.5B to 9B) evaluated by the same harness, each at the temperature it ships with. The Laya typed-decisions specialist,

	kev-4b	4B delta	4B + T	9B delta	9B + T	v2-9B
decision accuracy	0.665	0.800	0.800	0.774	0.774	0.783
ECE	0.091	0.162	0.009	0.149	0.021	0.140
confidence bias	−0.089	−0.162	+0.005	−0.149	+0.007	−0.140
Brier	0.467	0.327	0.276	0.350	0.310	0.334
NLL	0.819	0.576	0.460	0.612	0.511	0.585
KL(gold‖pred)	0.226	0.079	0.469†	0.093	0.302	—
AURC	0.195	0.071	0.069	0.087	0.085	0.079
coverage / acc. at conf. ≥0.9	.059 / .983	.130 / 1.00	.374 / .974	.120 / 1.00	.319 / .977	.133 / 1.00
score ECE	0.074	0.212	0.034	0.198	0.044	0.196
choice / noul ECE	.104 / .111	.149 / .107	.051 / .012	.131 / .104	.056 / .022	.135 / .084

Table 1: Held-out v1 test (400 cases, 1,945 decisions). "+T" is the five-fold group-disjoint cross-fit temperature ($T \approx 0.41$ at 4B, 0.45 at 9B). ECE 95% CIs: 4B $[0.143, 0.180]$ raw, $[0.009, 0.030]$ cross-fit; 9B $[0.131, 0.168]$, $[0.013, 0.039]$. †KL at the train-fit $T^{\star} = 0.35$ (ECE 0.032, NLL 0.464). v2-9B logits were not retained, so its temperature-scaled and KL columns are not computed. kev-4b is scored at its shipped $T = 2.41$.

a Laya checkpoint specialised to the same v1 training split (the Laya family is post-trained with proper-scoring RL rewards [3]). A zero-shot DeBERTa-v3 NLI model. The specialist and NLI predictions come from our `decisionbench` harness on all 2,000 decisions, and every other number from `kev.benchmark` on 1,945. Train-frequency prior accuracy is 0.484.

Metrics. Decision accuracy; top-label ECE (10 equal-width bins); confidence bias (mean confidence minus accuracy; negative means underconfident); Brier; NLL; KL(gold‖pred); AURC; coverage and accuracy at confidence ≥ 0.9; score MAE and RPS. Confidence intervals come from a case-level bootstrap (2,000 resamples), paired where two models are compared. Every model is a single training run, so seed variance is unmeasured.

5 Results

Proper-loss post-training improves accuracy, proper scores and selective prediction (Table 1). At 4B, accuracy rises from 0.665 to 0.800 (paired +0.135, 95% CI $[0.113, 0.157]$), Brier from 0.467 to 0.327, NLL from 0.819 to 0.576 and AURC from 0.195 to 0.071; KL from gold falls from 0.226 to 0.079. Coverage at confidence ≥ 0.9 doubles (0.059 to 0.130) with no errors inside the band. Gains hold in every workflow (agent-trace 0.568 to 0.754, customer service 0.706 to 0.822, invoice 0.692 to 0.840, security 0.700 to 0.786) and every primitive. The cold-started 9B delta reproduces every direction (0.774 accuracy, KL 0.093) and beats the released kev-9b by +0.057 $[0.035, 0.078]$. It trails the warm-started 4B by 0.026 $[0.010, 0.042]$. That gap is resolvable on these items, but it confounds scale with warm-start and has no seed replicate, so we read it as "no gain from scale under cold start", not as evidence against scale.

ECE regresses because the deltas are underconfident. Raw ECE rises from 0.091 to 0.162 at 4B and sits at 0.149 at 9B, with confidence bias equal to −ECE. In other words, the model is underconfident in aggregate, and Figure 2 shows it is underconfident in every bin at both scales. Score questions carry most of it (0.212 and 0.198); noul is mildest. More data does not remove it: v2-9B is still at −0.140 on the v1 test.

A temperature below one restores ECE, and the cost is KL. Cross-fit temperatures are stable across folds (4B: 0.40–0.43; 9B: 0.44–0.46) and bring ECE to 0.009 and 0.021, with Brier and NLL improving and accuracy fixed. Selective prediction becomes usable: at confidence ≥ 0.9 the 4B model now covers 37.4% of decisions at 97.4% accuracy, against 13.0% before. The train-fit $T^{\star} = 0.35$ at 4B gives 0.032, which is consistent. Per-primitive temperatures are similar to the global one at 9B (choice 0.52, noul 0.44, score 0.40; ECE 0.045, 0.007, 0.017), so one global T suffices in-domain. Sharpening, however, moves the output away from the gold distributions it was trained to match: KL rises from 0.079 to 0.469 at 4B and from 0.093 to 0.302 at 9B. A deployment has to choose between probabilities

model	acc.	ECE	bias	Brier	ECE after cross-fit T
kev-0.5b / 0.6b	.455 / .447	.114 / .227	+.114 / +.227	.680 / .727	—
kev-0.8b (shipped T=2.35)	0.458	0.047	+0.014	0.608	—
kev-8b (Qwen3)	0.638	0.112	+0.112	0.513	0.051 ($T \approx 1.7$)
kev-4b (shipped T=2.41)	0.665	0.091	−0.089	0.467	—
kev-9b (shipped T=2.30)	0.718	0.091	−0.091	0.414	—
NLI zero-shot*	0.531	0.081	+0.035	0.590	0.089 ($T \approx 1.46$)
Laya typed-decisions*	0.767	0.214	−0.214	0.400	0.018 ($T \approx 0.33$)
4B delta (ours)	0.800	0.162	−0.162	0.327	0.009 ($T \approx 0.41$)
9B delta (ours)	0.774	0.149	−0.149	0.350	0.021 ($T \approx 0.45$)

Table 2: All models on the v1 test. *`decisionbench` harness, 2,000 decisions; all others `kev.benchmark`, 1,945–1,955. Models that ship a temperature are not refit, because their recorded logits already include it. Hard-label-trained Kev models without a shipped temperature are overconfident (+); every soft-target-trained model is underconfident.

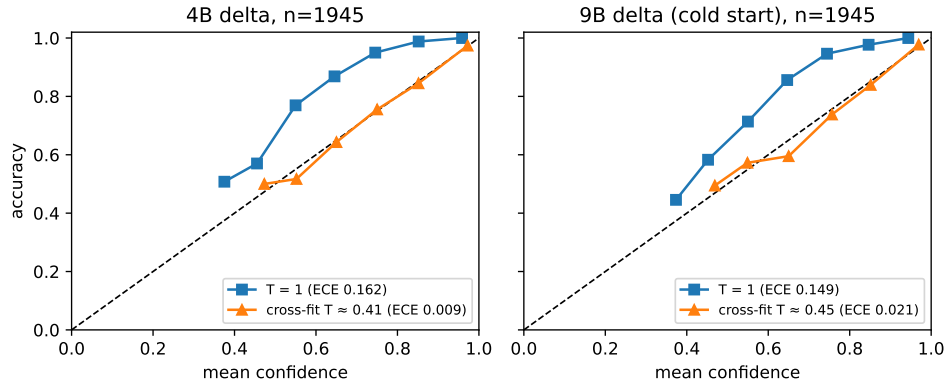


Figure 2: Reliability on the v1 test (accuracy vs. mean confidence, 10 bins, bins with ≥ 10 decisions). At $T = 1$ both deltas lie above the diagonal, meaning they are underconfident in every bin. The cross-fit temperature brings both onto it.

that match annotator disagreement and probabilities that match argmax hit rates. One temperature cannot give both.

Under shift, the sign flips. On the 6,264-decision v2 evaluation set, which is dominated by external sources with near-hard labels, v2-9B reaches 0.786 accuracy (choice 0.801, score 0.647, noul 0.783), but it is overconfident: mean confidence 0.893, bias +0.107 (choice +0.114, noul +0.145, score +0.035). At confidence ≥ 0.9 it covers 69% of decisions at 87.2% accuracy, against 100% on the v1 test. Any $T < 1$ fit in-domain would raise these confidences further. The direction of miscalibration is a property of how soft the evaluation labels are relative to the training targets, not a fixed property of the model.

Fair comparison with the baselines (Table 2). The Laya typed-decisions specialist is also underconfident on this data (bias −0.214). With the same cross-fit, its ECE falls from 0.214 to 0.018 ($T \approx 0.33$). After temperature scaling, our 4B delta and the specialist are therefore comparably calibrated (0.009 vs. 0.018), and our advantage is accuracy (0.800 vs. 0.767) and Brier (0.276 vs. 0.318). The NLI baseline is mildly overconfident ($T \approx 1.46$), and temperature does not improve its ECE (0.081 to 0.089). Among the released Kev models, accuracy grows with size except kev-8b (Qwen3 backbone; 0.638 < kev-4b's 0.665), and the sub-1B models cluster near 0.45.

6 Analysis and Limitations

Mechanism. The cross-entropy term is minimised when $p = g$, and the gold here has 0.75 nats of mean entropy, so the trained model reports spread-out distributions. The argmax

of those distributions is right more often than its probability says, because the teacher's majority label is usually the gold argmax. ECE scored against y therefore reads underconfidence [1, 11]. Two observations outside our own runs support this. The released Kev models, trained with hard-label cross-entropy, are overconfident before their shipped temperatures ($T > 2$), while our deltas from the same family need $T < 1$. And the independently trained Laya specialist on the same split is underconfident with a similar fitted temperature (0.33 vs. 0.41). This is not a controlled hard-versus-soft comparison: data, starting point and training stage all differ. Running that comparison on one backbone is the obvious next experiment.

What the temperature is fitting. Both of our fits minimise NLL against discrete labels, which is the estimator shown to understate annotator uncertainty [18, 7]. That is the right target if downstream thresholds act on argmax hit rates, and the wrong one if consumers read the probabilities as disagreement. The KL column of Table 1 quantifies that choice. Calibrators fit to the soft gold itself are the natural comparison and are untested here.

Limitations. (i) Every model is a single training run. Seed variance is unmeasured, and the 4B–9B gap confounds scale with warm-start. (ii) The train-fit temperature uses rows the model was trained on. Cross-fit temperatures on the test distribution are honest about held-out data from the same distribution, but they say nothing about shift, and the v2 evaluation shows that shift can reverse the sign. (iii) Tail calibration is unmeasured, because rare labels have single-digit training counts. (iv) The v2-9B logits were not retained, so its temperature analysis is limited to the sign of its bias. (v) The temperature hypothesis was formed by inspecting test-set reliability before the honest fits were run. We report the direction as post hoc and the validation as not. (vi) With 1,945 decisions, ECE values near 0.01 are close to the finite-sample noise floor [16], so 0.009 and 0.021 should not be ranked against each other.

7 Conclusion

Post-training a single-pass decision model with soft-target cross-entropy plus Brier and RPS terms improves accuracy, proper scores and selective prediction at 4B and 9B. It also makes the model underconfident against discrete labels, as the soft-label calibration literature predicts. One temperature below one repairs in-domain ECE without touching accuracy, but it trades away the distributional fidelity the loss bought, and it points the wrong way on external hard-label data, where the same model is overconfident. The practical rule is to fit temperatures per deployment distribution, against the label semantics the thresholds will use, and never ship a training-distribution temperature as a global constant.

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